

Junwei Huang

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EDUCATION

University of Macau — Ph.D. in Electrical and Computer Engineering	2018.08–2024.07
Advisors: Prof. Yan Lu and Prof. Chi-Seng Lam	
University of Electronic Science and Technology of China — B.Eng. in Microelectronics	2014.09–2018.06

ACADEMIC APPOINTMENTS

University of California, Berkeley, EECS — Postdoctoral Researcher	2025.09–present
Faculty Host: Prof. Robert Pilawa-Podgurski	
University of Macau, Institute of Microelectronics — Postdoctoral Fellow	2025.10–present
PI: Prof. Rui P. Martins and Prof. Sai-Weng Sin	
University of Macau, Institute of Microelectronics — Research Assistant	2024.10–2025.10
PI: Prof. Rui P. Martins	
Tsinghua University, Dept. of Electronic Engineering — Visiting Scholar	2024.11–2025.05
Faculty Host: Prof. Yan Lu	

RESEARCH SUMMARY

University of California, Berkeley, EECS 2025.09–present

- Leading research on vertical power delivery for next-generation AI-datacenter processors, spanning **in-package HV-IVR** design and **out-of-package board-level VRMs**, with emphasis on topology–package–system co-design and electro-thermal-mechanical modeling.

University of Macau, State Key Lab of Analog and Mixed-Signal VLSI 2018.08–2025.09

- Designed and taped out multiple hybrid SC/inductor IVRs in 65 nm CMOS and 180 nm BCD, covering the full IC design cycle from topology innovation, multi-loop control, circuit/layout implementation, to silicon measurement and publication.
- Demonstrated state-of-the-art efficiency (96.1% at 4-to-1.2 V), current density (1.46 A/mm² at 12:1), and transient response (63 mV droop @ 3.5 A load step) across hybrid converters with conversion ratios from 4:1 to 12:1.
- Pioneered a scalable multi-module parallel architecture with one-pin decentralized current balancing, reducing inter-module interconnect complexity — directly applicable to IVR arrays and vertical power delivery.
- **25 publications: 1 ISSCC highlight paper (1st author), 3 CICC 1st-author, 2 JSSC 1st-author, 1 TCAS-I 1st-author. Total: 6 ISSCC + 5 JSSC contributions.**

INVITED TALKS

1. Invited talk, **Google, Physical infrastructure - next-gen TPU power delivery Group**
2. Invited presenter, **IEEE PES/PELS Forum, UC Berkeley.**
3. Invited talk, **7th ICAC (2025) Chinese Chip Design Technology Workshop.**

AWARDS

PwrSoC 2023 Best Student Poster Award, International Workshop on Power Supply on Chip.

Akrostar Technology Academic Prize, academic year 2023/2024 (Top 3).

ACADEMIC SERVICE

Reviewer for IEEE JSSC, IEEE TIE, IEEE TCAS-II, and other IEEE journals.

PUBLICATIONS

Journal:

- [J1] **J. Huang**, Z. Tong, X. Mao, C.-S. Lam, R. P. Martins, Y. Lu, "A Fast-Slow Two-Module High-Power-Density DC-DC Solution With Transient and Efficiency Improvements," IEEE JSSC, 2025.
- [J2] **J. Huang**, Z. Tong, C.-S. Lam, X. Mao, R. P. Martins, Y. Lu, "A Multi-Path Inductor-First Inductor-on-Ground Switched-Capacitor Hybrid DC-DC Converter," IEEE JSSC, 2024.
- [J3] **J. Huang**, C.-S. Lam, Y. Lu, R. P. Martins, "A Symmetrical Double Step-Down Converter with Extended Voltage Conversion Ratio," IEEE TCAS-I, 2022.
- [J4] Z. Tong, **J. Huang**, X. Mao, R. P. Martins, Y. Lu, "A Double Pulse Overlapping Laser Diode Driver with Minimum 100-ps Pulse for LiDAR System," IEEE JSSC, 2024.
- [J5] Z. Tong, **J. Huang**, X. Mao, R. P. Martins, Y. Lu, "A Bidirectional USB Power Delivery Voltage-Regulating Cable," IEEE JSSC, 2024.
- [J6] W. Yang, Z. Tong, **J. Huang**, R. P. Martins, Y. Lu, "A Bi-Directional Dual-Path Boost-Buck Hybrid Converter for High-Voltage Power Transmission Delivery Cable in Humanoid Robots," IEEE JSSC, 2025.
- [J7] Y. Lu, **J. Huang**, Z. Tong, T. Hu, W.-L. Zeng, M. Huang, X. Mao, G. Cai, "An Overview of Hybrid DC-DC Converters: From Seeds to Leaves," IEEE OJ-SSCS, 2024.
- [J8] Y. Lu, G. Cai, **J. Huang**, "Favorable basic cells for hybrid DC-DC converters," J. Semiconductors, 2023.

Conference:

- [C1] **J. Huang**, X. Mao, Z. Tong, Z. Yu, W. Yang, C.-S. Lam, R. P. Martins, Y. Lu, "A 20MHz-1MHz Dual-Loop Non-Uniform-Multi-Inductor Hybrid DC-DC Converter with Specified Inductor Current Allocation and Fast Transient Response," **ISSCC 2025. (Highlight Paper)**
- [C2] **J. Huang**, Z. Tong, X. Mao, C.-S. Lam, R. P. Martins, Y. Lu, "A Fast-Slow Two-Module DC-DC Solution with Transient and Efficiency Improvements for 2.5D/3D Integration," CICC 2024.
- [C3] **J. Huang**, Z. Tong, Y. Lu, C.-S. Lam, R. P. Martins, "A 5V-to-0.5V Inductor-First Inductor-on-Ground Switched Capacitor Multi-Path Hybrid DC-DC Converter," CICC 2023.
- [C4] **J. Huang**, X. Mao, Z. Tong, Z. Yu, S. Han, C.-S. Lam, R. P. Martins, Y. Lu, "A Scalable Distributed 12-to-1V Fast-Slow Hybrid DC-DC Combo with Fast-Transient Bucklets and One-Pin Current Balancing," CICC 2026.
- [C5] Z. Tong, **J. Huang**, Y. Lu, R. P. Martins, "A 42W Reconfigurable Bidirectional Power Delivery Voltage-Regulating Cable," **ISSCC 2023. (Highlight Paper)**
- [C6] Z. Zhu, **J. Huang**, Z. Song, Z. Yu, S. Han, S.-W. Sin, Y. Lu, "An Inductor-at-Middle Hybrid Buck Converter with Shared Power-Signal Path for Distributed Vertical Power Delivery," **ISSCC 2026.**
- [C7] Z. Yu, Z. Tong, **J. Huang**, J. Yin, Y. Lu, "A Compact Dual-Capacitor Relay SPT Supply Modulator with Overshoot-Free Adaptive On-Time Control for 5G FR2 CMOS PA," **ISSCC 2026.**
- [C8] Z. Tong, Z. Yu, **J. Huang**, X. Mao, B. Wicht, R. P. Martins, Y. Lu, "HOOP: A Scalable Hybrid DC-DC Converter Ring for High Performance Computing," **ISSCC 2025.**
- [C9] W. Yang, Z. Tong, **J. Huang**, R. P. Martins, Y. Lu, "A Bi-Directional Dual-Path Boost-48V-Buck Hybrid Converter for High-Voltage Power Transmission Cable in Light-Weight Humanoid Robots," **ISSCC 2025.**
- [C10] Z. Yu, **J. Huang**, Z. Tong, Z. Zhu, Y. Lu, "An Output-Rail-Scalable Hybrid Buck Converter with Globally Adaptive Frequency Control Achieving 98.0% Peak Efficiency and 490 mW/mm³ Power Density," CICC 2026.
- [C11] S. Han, **J. Huang**, F. Song, Z. Tong, S.-W. Sin, H. Jiang, X. Wu, T. Ren, Y. Lu, "A 94.1% Peak Efficiency Coupled-Inductor Boost Converter with Small Inductor Current Ripple and Fast Transient Response," CICC 2026.
- [C12] Z. Song, Z. Tong, **J. Huang**, Z. Zhu, S.-W. Sin, Y. Lu, "A 6V-to-1V ReSC-Hybrid Two-Stage Merged Buck Converter with an Ultra-Small Inductor for VPD," CICC 2026.
- [C13] X. Mao, **J. Huang**, Z. Tong, R. P. Martins, Y. Lu, "A Quad-Output Hybrid Buck Converter with 8-Inductor Helping One Spot from All Quarters for Multi-Core XPU's," CICC 2024.
- [C14] Z. Tong, W. Yang, S. Han, **J. Huang**, X. Mao, Y. Lu, "Where is the Inductor: A Review and Comparison of the Hybrid DC-DC Buck Topologies," CICC 2025.

- [C15] Z. Yu, **J. Huang**, Z. Tong, M. Huang, Y. Lu, "An Always Dual-Path Hybrid DC-DC Converter with Multiphase Interleaving Switched-Capacitor Cell Obtaining 45% Output Ripple Reduction," ISCAS 2025.
- [C16] F. Luo, **J. Huang**, M. Huang, Y. Lu, "A 12V-Input 1.8V-0.8V-Output Multiple-Output Hybrid Buck DC-DC Converter with a Shared Flying Capacitor," ISCAS 2025.
- [C17] Y. Hu, **J. Huang**, M. Huang, Y. Lu, "A 5V-to-0.8V Inductor-First 2L2C Multi-Path Hybrid DC-DC Converter," ISCAS 2025.